

```

# Data handling
import pandas as pd
import numpy as np

# Text preprocessing
import string
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split

# Model
from sklearn.naive_bayes import MultinomialNB

# Evaluation
from sklearn.metrics import accuracy_score, precision_score, recall_score, f

```

```

!wget -O spam.csv https://raw.githubusercontent.com/mohitgupta-omg/Kaggle-SM

df = pd.read_csv("spam.csv", encoding='latin-1')

df = df[['v1', 'v2']]
df.columns = ['label', 'message']

print(df.head())
print("Dataset size:", df.shape)
print(df['label'].value_counts())

```

```

--2026-02-18 05:58:07-- https://raw.githubusercontent.com/mohitgupta-omg/Kaggle-SM
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.10
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.1
HTTP request sent, awaiting response... 200 OK
Length: 503663 (492K) [application/octet-stream]
Saving to: 'spam.csv'

```

```

spam.csv          100%[=====>] 491.86K  --.-KB/s    in 0.04s

```

```

2026-02-18 05:58:07 (12.8 MB/s) - 'spam.csv' saved [503663/503663]

```

```

      label      message
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...
Dataset size: (5572, 2)
label
ham      4825
spam      747
Name: count, dtype: int64

```

```

import nltk
from nltk.corpus import stopwords
nltk.download('stopwords')

```

```

stop_words = set(stopwords.words('english'))

def preprocess(text):
    text = text.lower()
    text = ''.join([char for char in text if char not in string.punctuation])
    words = text.split()
    words = [word for word in words if word not in stop_words]
    return " ".join(words)

df['clean_message'] = df['message'].apply(preprocess)

```

```

[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.

```

```

vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(df['clean_message'])
y = df['label']

print("Feature matrix shape:", X.shape)
print("Sample features:", vectorizer.get_feature_names_out()[:10])

```

```

Feature matrix shape: (5572, 9376)
Sample features: ['008704050406' '0089my' '0121' '01223585236' '01223585334'
'02' '020603' '0207' '02070836089']

```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

```

```

model = MultinomialNB()
model.fit(X_train, y_train)

```

```

print(model.get_params())

```

```

{'alpha': 1.0, 'class_prior': None, 'fit_prior': True, 'force_alpha': True}

```

```

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, pos_label='spam'))
print("Recall:", recall_score(y_test, y_pred, pos_label='spam'))
print("F1 Score:", f1_score(y_test, y_pred, pos_label='spam'))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))

```

Accuracy: 0.9659192825112107
Precision: 1.0
Recall: 0.7466666666666667
F1 Score: 0.8549618320610687
Confusion Matrix:

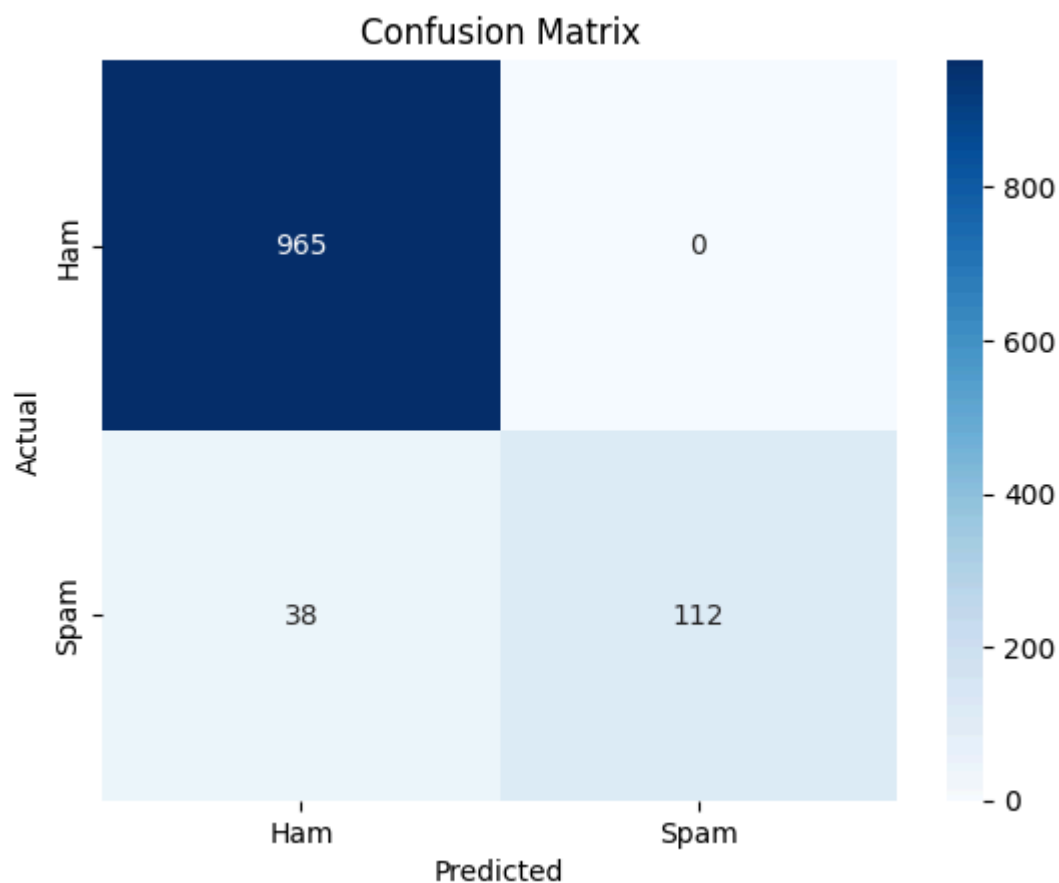
		precision	recall	f1-score	support
	ham	0.96	1.00	0.98	965
	spam	1.00	0.75	0.85	150
accuracy				0.97	1115
macro avg		0.98	0.87	0.92	1115
weighted avg		0.97	0.97	0.96	1115

```
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Ham', 'Spam'],
            yticklabels=['Ham', 'Spam'])

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



Start coding or generate with AI.